

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A 50 g ball falls vertically from a height of 2.5 m onto a hard surface.
After the bounce, it rises to a height of 1.8 m.

(a)

Calculate the speed of the ball just before it hits the surface.

Use energy conservation:

$$E_p = E_k \quad \Rightarrow \quad mgh = \frac{1}{2}mv^2$$

Mass cancels:

$$gh = \frac{1}{2}v^2 \quad \Rightarrow \quad v = \sqrt{2gh}$$

$$v = \sqrt{2 \times 9.8 \times 2.5} = \sqrt{49} = 7.0 \text{ m/s}$$

✔ *Revision tip:*

To find speed from height, use:

$$v = \sqrt{2gh}$$

It avoids needing the time of fall.

(b)

Calculate the speed of the ball as it leaves the surface after the bounce.

Again use:

$$v = \sqrt{2gh}$$

$$v = \sqrt{2 \times 9.8 \times 1.8} = \sqrt{35.28} = 5.94 \text{ m/s}$$

✔ *Revision tip:*

This is the same method — but now the **maximum height after the bounce** gives the **initial speed upward**.

(c)

Calculate the percentage of kinetic energy lost during the bounce.

Initial KE before impact:

$$E_{k1} = \frac{1}{2}mv_1^2 = \frac{1}{2} \times 0.050 \times 7.0^2 = 1.225 \text{ J}$$

Final KE after bounce:

$$E_{k2} = \frac{1}{2}mv_2^2 = \frac{1}{2} \times 0.050 \times 5.94^2 = 0.882 \text{ J}$$

Energy lost:

$$\Delta E = 1.225 - 0.882 = 0.343 \text{ J}$$

Percentage lost:

$$\frac{0.343}{1.225} \times 100 = 28\%$$

✔ *Revision tip:*

Use kinetic energy values to measure how **efficient** the bounce is.

If no energy were lost, the ball would rise to its original height!